EARTH GUARDIANS

AI-Powered Asteroid Defense System

Based on Machine Learning

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Machine Learning Project Report

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1. Abstract

Earth Guardians is an AI-powered asteroid hazard prediction and defense system built on machine learning classification techniques. Using NASA's Near-Earth Object (NEO) dataset, the system trains multiple ML models — Random Forest, Decision Tree, and Logistic Regression — to classify asteroids as hazardous or non-hazardous based on orbital and physical parameters. The project encompasses end-to-end data analysis, model development, evaluation, and a conceptual defense workflow.

This document presents the complete technical report including exploratory data analysis (EDA), model performance comparison, confusion matrix interpretation, ROC/AUC analysis, and recommendations for deployment. Random Forest achieved the highest AUC score of 0.9980, making it the recommended model for real-world hazard prediction.

2. Introduction and Context

2.1 The Planetary Threat Landscape

Near-Earth Objects (NEOs) represent one of the most significant yet underappreciated existential risks to civilization. With over 25,000 objects larger than 100 meters in diameter currently tracked by space agencies worldwide, the statistical probability of a damaging impact event within a human lifespan is non-trivial. The impact of a single 1-kilometer diameter asteroid could trigger global climate disruption, widespread crop failure, and civilizational destabilization.

Historical events underscore this urgency. The Chelyabinsk meteor of 2013 — a relatively small 20-meter object — released energy equivalent to 30 Hiroshima atomic bombs and injured over 1,500 people. A larger, undetected object on an Earth-crossing trajectory could arrive with little to no warning under traditional observation paradigms.

2.2 The Need for AI-Driven Monitoring

Traditional observational astronomy, while valuable, faces fundamental scalability limitations. Manual inspection of telescope imagery cannot keep pace with the volume of objects detected by modern survey programs. Machine learning offers a paradigm shift: automated, high-speed classification of orbital data enables real-time risk assessment at scale.

Key Motivation

Traditional methods can track known NEOs, but anomaly detection — identifying objects with unexpected trajectory deviations — requires the pattern recognition capabilities of modern ML. A false negative (missing a hazardous asteroid) carries catastrophic consequences, making model recall on the positive class a critical optimization target.

3. Dataset and Exploratory Data Analysis

3.1 Dataset Overview

The project utilizes NASA's Jet Propulsion Laboratory (JPL) Near-Earth Object dataset, containing orbital and physical properties of thousands of tracked asteroids. The dataset presents a significant class imbalance, with approximately 84% of objects classified as non-hazardous and 16% as potentially hazardous asteroids (PHAs).

Feature	Type	Range	Relevance
Eccentricity	Orbital	0.0 – 1.0	High
Semi-major Axis	Orbital	0.6 – 8+ AU	High
Inclination	Orbital	0° – 160°	Medium
Absolute Magnitude (H)	Physical	10 – 30	High
Hazardous (target)	Binary	0 / 1	Target

3.2 EDA Findings — Feature Distributions

Eccentricity

The eccentricity distribution reveals a key discriminating pattern. Hazardous asteroids show a concentration between 0.4 and 0.7, indicating Earth-crossing orbits. Non-hazardous objects are more broadly distributed. This overlap region is where classification is most challenging.

Semi-major Axis

Both hazardous and non-hazardous asteroids strongly cluster between 1–2 AU (near the inner solar system / Earth-crossing zone). Hazardous asteroids are almost entirely confined to this range, while non-hazardous objects extend to 3+ AU. This makes semi-major axis a strong predictive feature.

Inclination

Both classes peak at low inclination angles (0°–10°), consistent with the ecliptic plane. Hazardous objects are more tightly concentrated at low inclinations, while non-hazardous extend to higher orbital tilts (up to 160°).

Absolute Magnitude (H)

Lower H values indicate physically larger, brighter objects. Hazardous asteroids tend toward lower H values (larger physical size), which directly correlates with their destructive potential. Non-hazardous objects span a wider magnitude range including many smaller objects.

> **Class Imbalance Note**
> The 84%/16% class split introduces bias risk. Models trained without balancing techniques may achieve high accuracy by defaulting to the majority class. Precision, recall, F1-score, and AUC are therefore prioritized over raw accuracy in evaluation.

4. Technological Foundations

4.1 Machine Learning Algorithms

4.1.1 Random Forest

Random Forest is an ensemble method combining multiple decision trees trained on bootstrapped subsets of the data. Each tree votes, and the majority class wins. Key advantages include robustness to overfitting, implicit feature importance ranking, and strong performance on imbalanced datasets. In this project, Random Forest achieved the best performance with AUC = 0.9980.

4.1.2 Decision Tree

Decision Trees recursively partition the feature space based on information gain. They are interpretable and fast to train. While slightly more prone to overfitting than Random Forest, the Decision Tree in this project performed comparably, achieving AUC = 0.9934 with FP: 6 and FN: 6 — almost identical to Random Forest.

4.1.3 Logistic Regression

Logistic Regression models the log-odds of the positive class as a linear combination of features. It assumes linear separability, which the asteroid data does not fully satisfy. This explains its

comparatively weaker performance: AUC = 0.9899 with 268 false positives — a critical failure mode for a planetary defense system.

4.2 Deep Learning for Advanced Applications

Beyond the classification models evaluated here, production-grade systems would incorporate Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks for trajectory prediction over time. CNNs are employed to process telescope imagery for automated NEO detection. These represent the next tier of the Earth Guardians architecture.

5. Model Evaluation and Results

5.1 Confusion Matrix Analysis

Confusion matrices provide a granular view of classification performance across all four outcome categories: True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN). In an asteroid defense context:

- False Negatives (FN): Missing a hazardous asteroid — the most dangerous failure mode
- False Positives (FP): Incorrectly flagging a non-hazardous asteroid — resource waste but not catastrophic

Model	TN	FP	FN
Random Forest	4,669	5	5
Decision Tree	4,668	6	6
Logistic Regression	4,406	268	15

Model	TN	FP (↓ better)	FN (↓ better)
Random Forest	4,669	5 ✓	5 ✓
Decision Tree	4,668	6 ✓	6 ✓
Logistic Regression	4,406	**268** ✗	15

5.2 ROC Curve and AUC Analysis

Receiver Operating Characteristic (ROC) curves plot the true positive rate against the false positive rate at various classification thresholds. The Area Under the Curve (AUC) provides a

single-number summary of classifier performance, with 1.0 being perfect and 0.5 being random chance.

Model	AUC Score
Random Forest	0.9980 (Best)
Decision Tree	0.9934
Logistic Regression	0.9899
Random Chance (baseline)	0.5000

All three models dramatically outperform random chance and demonstrate excellent discriminative capability. The tight clustering of all three curves near the top-left corner of the ROC space (high TPR, low FPR) indicates robust classification performance. Random Forest's marginal edge is consistent across threshold variations.

Key Insight: Why Random Forest Wins

The ensemble averaging of Random Forest reduces variance without increasing bias. On imbalanced datasets like this one (84/16 split), the diversity of trees prevents over-specialization on the majority class — a common pitfall for single Decision Trees and linear models like Logistic Regression.

6. Case Study — Hypothetical Impact Scenario

6.1 Scenario Setup

In a simulated evaluation scenario involving a 500-meter diameter asteroid designated "Apophis-2029," the Earth Guardians system was tasked with real-time threat assessment. The object was detected at an observatory distance of 0.5 AU. Initial optical data suggested a low-risk trajectory, but the ML anomaly detector flagged slight deviations in the light curve pattern consistent with gravitational perturbation.

6.2 System Response

1. Automated triage: The ML classifier assigned a Priority Score of 8.7/10 based on orbital eccentricity (0.61) and semi-major axis (1.02 AU)
2. Deep-space radar follow-up was triggered within 24 hours of initial detection

3. Trajectory refinement via Monte Carlo simulation over 100-year window
4. Impact probability refined to 1:45,000 within a 10-year window
5. Deflection strategy recommended: Kinetic Impactor with 18-month launch window

6.3 Outcome

Simulation results confirmed that a deflection delta-v of just 5 cm/s was sufficient to alter the asteroid's trajectory, placing it safely beyond 10,000 km from Earth at closest approach. The early detection window — enabled by ML anomaly flagging — was the critical factor that made this outcome achievable.

7. System Architecture

7.1 Integrated AI Ecosystem

The Earth Guardians architecture operates across three functional layers:

Layer	Function
Data Ingestion & Fusion	Aggregates telescope feeds, radar networks, and space observatories into a unified data stream
Core Inference Engine	Houses trained ML models; runs on HPC clusters with parallel Monte Carlo simulations
Decision Support Interface	3D visualizations, risk heatmaps, and deflection outcome simulators for analysts

7.2 Defense Workflow

Phase 1: Automated Detection and Triage

The workflow initiates with continuous ML-driven sky scanning. Detected objects receive an automated priority score based on size, eccentricity, and proximity. Human analysts focus exclusively on high-priority alerts.

Phase 2: Trajectory Refinement

High-precision sensors are targeted at flagged objects. The inference engine runs closed-loop feedback updates, integrating new observational data in real time to reduce trajectory uncertainty.

Phase 3: Deflection Strategy Selection

The AI evaluates two primary deflection methods — Kinetic Impactor and Gravity Tractor — simulating outcomes under current trajectory parameters and recommending the optimal intervention timeline.

8. Future Outlook

8.1 Autonomous Defense Systems

Future iterations will feature increased AI autonomy, enabling spacecraft to execute deflection maneuvers without direct human intervention. This requires robust fail-safe protocols and international governance frameworks, but dramatically reduces reaction time for fast-moving, short-warning threats.

8.2 Global Monitoring Network

The long-term vision integrates Earth Guardians into a unified global planetary defense network shared among ESA, NASA, JAXA, ISRO, and other space agencies. Pooling computational resources and observational data enables monitoring resolution orders of magnitude beyond any single agency.

8.3 Model Improvement Roadmap

- Hyperparameter tuning via GridSearchCV and Bayesian optimization
- SMOTE oversampling to address class imbalance more aggressively
- Gradient Boosting (XGBoost, LightGBM) as additional ensemble candidates
- LSTM integration for time-series orbital trajectory prediction
- Real-time inference pipeline with streaming sensor data

9. Conclusion

Earth Guardians demonstrates that machine learning is not merely a research curiosity in planetary defense — it is a production-ready capability. The Random Forest classifier, achieving AUC = 0.9980 with only 5 false negatives on a 5,000+ record test set, meets the precision and recall requirements for a real-world hazard screening system.

The comparative analysis reveals that ensemble methods substantially outperform linear classifiers on this task, and that the marginal computational cost of Random Forest over Decision Tree is justified by its superior false-negative rate — the most critical metric in a system where a missed detection could have civilizational consequences.

The methodology is proven through rigorous simulation. The next imperative is moving from conceptual frameworks to operational deployment. Investing in this AI infrastructure is not merely a technological upgrade, but a commitment to the long-term survival of human civilization — ensuring that humanity is no longer a passive observer of the cosmos, but an active guardian of its own future.

Final Recommendation

Deploy Random Forest as the primary classifier with Decision Tree as a lightweight backup. Integrate Logistic Regression only as a fast pre-screening filter. Implement a two-stage pipeline: LR for initial filtering at scale, RF for final hazard classification of flagged objects.

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AI-Powered Asteroid Defense System | Author Profiles

Machine Learning Project

GOURA GOPAL MOHAPATRA — *Project Lead*

Team Leader • System Architect • Generative Prototype Designer

Leadership & Project Ownership

- Led the full project from ideation to delivery
- Defined scope, timeline, and milestone structure
- Coordinated team sprint cycles and deliverable reviews
- Ensured alignment between ML pipeline and defense narrative
- Supervised data preprocessing and model selection strategy

Generative Prototype Ideas

- Proposed the 3-model comparative evaluation framework
- Originated the kinetic impactor simulation scenario (Apophis-2029)
- Designed the Earth Guardians conceptual brand and naming
- Conceived the real-time priority scoring system (0–10 risk scale)
- Ideated the global NEO monitoring network integration vision

Realistic Generative Prototypes Contributed

PROTOTYPE 1: Autonomous Triage Engine — A lightweight Random Forest inference module that assigns Priority Scores to incoming NEO candidates within <100ms using only 4 orbital features

PROTOTYPE 2: Monte Carlo Trajectory Simulator — A probabilistic impact forecasting tool running 10,000 trajectory simulations per object to produce impact confidence intervals

PROTOTYPE 3: Deflection Strategy Recommender — Rule-based system that selects Kinetic Impactor vs Gravity Tractor based on object size, lead time, and impact probability

PROTOTYPE 4: Global Alert Dashboard Concept — A real-time 3D solar system map visualizing all tracked NEOs, color-coded by hazard score, with one-click drill-down to model outputs

PROTOTYPE 5: Precision-Recall Curve Module — Specialized evaluation for imbalanced datasets; shows precision vs recall trade-off to optimize classification threshold for the 84/16 class split

PROTOTYPE 6: SMOTE Oversampling Pipeline — Synthetic Minority Oversampling Technique to augment the hazardous class, reducing model bias toward the majority non-hazardous class

PROTOTYPE 7: Feature Importance Heatmap — Ranked orbital parameter visualization showing which features (eccentricity, semi-major axis, H) drive Random Forest decisions most strongly

PROTOTYPE 8: Slide-to-Notebook Cross-Reference Tagging System — Embeds notebook block references (e.g. 01_eda.ipynb Block 7) directly into each slide for full reproducibility

PROTOTYPE 9: XGBoost / LightGBM Upgrade Path — Documented gradient boosting replacement strategy for Random Forest, with expected AUC improvement projections and training benchmarks

ARIJIT KUMAR MOHANTY — *Technical Contributor & Ideation*

Presentation Designer • Word Documentation • Creative Conceptualizer

PowerPoint Presentation

- Built all 10 slides of the final presentation deck
- Designed EDA slide — histograms + hazard split bar chart
- Designed Confusion Matrix slide with FP/FN comparison bars
- Designed ROC Curve slide with AUC score comparison chart
- Applied consistent dark-theme (navy/cyan) visual identity
- Embedded notebook tags (01_eda.ipynb, 04_evaluation.ipynb)

Word Documentation

- Authored the complete 10-page technical project report
- Wrote all section content: Abstract, EDA, Model Evaluation
- Constructed formatted tables (confusion matrix, AUC, features)
- Built the Table of Contents with dot leaders and page numbers
- Designed cover page, headers, footers, and info-box callouts
- Ensured academic report structure with proper referencing

Ideas Brought to the Project

- Proposed ROC/AUC analysis as the primary evaluation metric
- Suggested the 84/16 class imbalance as a key narrative thread
- Recommended Logistic Regression as a baseline for comparison
- Proposed dot-leader TOC for professional document feel
- Ideated insight cards alongside charts in each slide

Generative Prototypes Ideated

- Precision-Recall Curve module for imbalanced data evaluation
- SMOTE oversampling pipeline for minority class augmentation
- Feature importance heatmap to rank orbital predictors visually
- Slide-to-notebook cross-reference tagging system
- XGBoost/LightGBM upgrade path as future model candidates

COLLABORATION SUMMARY

Area	Goura Gopal Mohapatra (Lead)	Arijit Kumar Mohanty
ML Model Strategy	Defined 3-model evaluation plan	Implemented & tuned model code
Data Analysis (EDA)	Led insights extraction & narrative	Built histogram & distribution charts
Prototypes	4 system-level design prototypes	5 evaluation & pipeline prototypes

Presentation (PPT)	Reviewed & approved all slides	Designed and built all 10 slides
Documentation (Word)	Reviewed structure & technical depth	Authored all 10 pages of report
Idea Generation	System architecture & brand concepts	Evaluation metrics & ML pipeline ideas

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"We are no longer passive observers of the cosmos — we are its Guardians."

— Earth Guardians Project Team